

Assessment Schedule – PRACTICE ASSESSMENT A

Calculus: Apply differentiation methods in solving problems (91578)

Evidence Statement

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
ONE (a)	$f'(x) = \frac{2}{3} \times 3 \times (3x - 2)^{-\frac{1}{3}}.$	• Correct derivative.		
(b)	$\frac{dy}{dx} = (4x - 5) \cos(4x) - 4 \sin(4x) (2x^2 - 5x + 3).$ When $x = \frac{\pi}{4}$: $\frac{dy}{dx} = (4x - 5) \cos(\pi) - 4 \sin(\pi) (2x^2 - 5x + 3).$ $\frac{dy}{dx} = 5 - \pi.$ <i>Accept 1.8(58).</i>	• Correct gradient, with evidence of derivative.		
(c)	$f'(x) = \frac{(x - 1)(2x - 2) - (x^2 - 2x - 3)}{(x - 1)^2},$ $= \frac{x^2 - 2x + 5}{(x - 1)^2}.$ $\therefore$ parallel, $\therefore$ gradient of the normal = gradient of $y = -\frac{1}{2},$ Hence the gradient of the tangent = 2. Solve $f'(x) = 2,$ $x^2 - 2x - 3 = 0,$ $(x - 3)(x + 1) = 0,$ $x = 3$ or $x = -1.$ $\therefore x > 1,$ $\therefore x = 3$ is the only solution.	• Correct derivative.	• Correct x -coordinate found, with evidence of derivative.	
(d)	$\frac{dy}{dt} = 4t - 1, \frac{dx}{dt} = 2t \ln(2t) + \frac{2t^2}{2t},$ $\frac{dy}{dx} = \frac{4t - 1}{2t \ln(2t) + t}.$ At the stationary point, $\frac{dy}{dx} = 0,$ $4t - 1 = 0, t = \frac{1}{4},$ When $t = \frac{1}{4}, y = -\frac{1}{8},$ the equation of this tangent $y = -\frac{1}{8}.$ The curve crosses the y -axis at $x = 0,$ $t^2 \ln(2t) = 0, t = \frac{1}{2},$ When $t = \frac{1}{2}, x = 0, y = 0, \frac{dy}{dx} = 2,$ the equation of this tangent $y = 2x.$ To find the intersection, $-\frac{1}{8} = 2x \Rightarrow x = -\frac{1}{16}.$ The coordinates of the point $(-\frac{1}{16}, -\frac{1}{8}).$	• Correct expression for $\frac{dy}{dx}.$	• Correct coordinates found, with evidence of derivative.	

(e)	$f'(x) = 6x \sec^2(3x^2) - 6x,$ $f''(x) = 6 \sec^2(3x^2) +$ $72x^2 \sec^2(3x^2) \tan(3x^2) - 6.$ The gradient of tangent = $f''(x)$. At $x = \sqrt{\frac{\pi}{3}},$ $f''\left(\sqrt{\frac{\pi}{3}}\right) = 6 \sec^2(\pi) +$ $24\pi \sec^2(\pi) \tan(\pi) - 6 = 0,$ $\therefore m_{\text{tangent}} = 0.$ Hence the tangent to the graph at $x = \sqrt{\frac{\pi}{3}}$ is parallel to the y-axis. Since the normal to the graph at $x = \sqrt{\frac{\pi}{3}}$ is perpendicular to this tangent, the line L is parallel to the y-axis.	• Correct expression for $f'(x)$.	• Correct expression for $f'(x)$ AND $f''(x)$.	E7 (T1) Justifies that the tangent is parallel to the x-axis, with evidence of a calculus method OR Correct solution, but with one minor error. E8 (T2) Complete proof, with evidence of a calculus method.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
TWO (a) (i) (ii) (iii)	$-5 < x < -4, x = -2, x = 7$ $-4 < x < 0, 5 < x < 7$ Does not exist	• TWO out of three responses correct.		
(b)	$f'(x) = \frac{2e^{2x}(x+2)^2 - 2(x+2)e^{2x}}{(x+2)^4},$ When $x = 0, f'(0) = \frac{8-4}{16} = \frac{1}{4}.$ Since $f'(0) > 0$, the function is increasing at $x = 0$.	• Correct decision of increasing, with correct derivative.		
(c)	$\frac{dx}{dt} = 3t^2 + 3,$ $\frac{dy}{dt} = 6t^5 - 12t^3 - 18t.$ $\frac{dy}{dx} = \frac{6t(t^4 - 2t^2 - 3)}{3(t^2 + 1)} = \frac{6t(t^2 + 1)(t^2 - 3)}{3(t^2 + 1)},$ $= 2t(t^2 - 3) = 2t^3 - 6t.$ $\frac{d^2y}{dx^2} = \frac{d}{dt}(2t^3 - 6t) \times \frac{1}{3t^2 + 3},$ $= \frac{6t^2 - 6}{3t^2 + 3} = \frac{2(t^2 - 1)}{t^2 + 1}.$ To find point(s) of inflection, $\frac{d^2y}{dx^2} = 0, 2(t^2 - 1) = 0, t = \pm 1.$ When $t = 1, x = 4, y = -11.$ When $t = -1, x = -4, y = -11.$ The two points of inflection: (4, -11) and (-4, -11).	• Correct expression for $\frac{dy}{dx}.$	• Correct coordinates for points of inflection, with evidence of derivative.	
(d)	$\frac{dy}{dx} = 3e^{3x} \sin x + e^{3x} \cos x.$ $\frac{d^2y}{dx^2} = 9e^{3x} \sin x + 3e^{3x} \cos x + 3e^{3x} - e^{3x} \sin x,$ $= 8e^{3x} \sin x + 6e^{3x} \cos x.$ Given $\frac{d^2y}{dx^2} + ky = 6\frac{dy}{dx}$ Solving $8e^{3x} \sin x + 6e^{3x} \cos x + k(e^{3x} \sin x) = 6(3e^{3x} \sin x + e^{3x} \cos x),$ $(k - 10)(e^{3x} \sin x) = 0,$ $\therefore e^{3x} \sin x \neq 0$ for all $x,$ $\therefore k = 10.$	• Correct expression for $\frac{dy}{dx}.$	• Correct value of k found, with evidence of derivative.	
(e)	$\frac{dy}{dx} = \frac{e^x - e^{-x}}{e^x + e^{-x}},$ $\frac{d^2y}{dx^2} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2},$ $= \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{(e^x + e^{-x})^2},$ $= \frac{4}{(e^x + e^{-x})^2}.$ $\therefore 4 \neq 0$ and $(e^x + e^{-x})^2 \neq 0, \frac{d^2y}{dx^2} \neq 0,$ $\therefore$ The curve has NO points of inflection. Since $4 > 0$ and $(e^x + e^{-x})^2 > 0,$ $\frac{d^2y}{dx^2}$ is therefore positive and the curve is concave up everywhere.	• Correct expression for $\frac{dy}{dx}.$	• Correct expression for $\frac{dy}{dx}$ AND $\frac{d^2y}{dx^2}.$	E7 (T1) Correct solution, but with one minor error. E8 (T2) Correct conclusion with full justification and evidence of a calculus method.

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
THREE (a)	$f'(x) = 4x \cdot \cot x^3 + 2x^2 \cdot 3x^2 \cdot (-\operatorname{cosec}^2(x^3))$.	• Correct derivative.		
(b)	$T = \frac{30t}{e^{\frac{t}{6}}}$, $T' = \frac{e^{\frac{t}{6}} \times 30 - 30t(\frac{1}{6}e^{\frac{t}{6}})}{\left(e^{\frac{t}{6}}\right)^2}$. To find the turning point, let $T' = 0$. $e^{\frac{t}{6}} \times 30 - 30t\left(\frac{1}{6}e^{\frac{t}{6}}\right) = 0$, $30 - 5t = 0$, $t = 6$ (in the sixth minute).	• Correct time t found, with evidence of derivative.		
(c)	Method 1 $\tan(\theta) = \frac{x^2 + \frac{4}{x}}{x} = x + \frac{4}{x^2}$. Let $f(x) = x + \frac{4}{x^2}$, Since θ is an acute angle ($0^\circ < \theta < 90^\circ$) inside a right-angled triangle, the x value that minimises $f(x)$ will also minimise $\tan(\theta)$. $f'(x) = 1 - 8x^{-3}$. To find minimum, $f'(x) = 0$. Solving the equation $1 - \frac{8}{x^3} = 0$, $x = 2$. OR Method 2 $\frac{d\theta}{dx} \cdot \sec^2(\theta) = 1 - 8x^{-3}$, Hence $\frac{d\theta}{dx} = \frac{1-8x^{-3}}{\sec^2(\theta)}$. To find minimum, $\frac{d\theta}{dx} = 0$. Solving the equation $\frac{1-8x^{-3}}{\sec^2(\theta)} = 0$, where $\sec^2(\theta) \neq 0$, $1 - \frac{8}{x^3} = 0$, $x = 2$. AND $f(2) = 3$, so $\tan(\theta) = 3$, $\theta = \tan^{-1}(3) \approx 1.249$ radians or 71.565° .	• Correct expression for $f'(x)$ AND $f'(x) = 0$ OR Correct expression for $\frac{d\theta}{dx}$ AND $\frac{d\theta}{dx} = 0$.	• Correct x value AND angle θ found, with evidence of derivative AND Recognises that the x value that minimises $f(x)$ will also minimise $\tan(\theta)$ since $0^\circ < \theta < 90^\circ$ if solved by Method 1 OR $\sec^2(\theta) \neq 0$ if solved by Method 2	
(d)	Surface area $S = 2\pi r^2 + 2\pi rh$, given that $h = 2r$, $S = 2\pi r^2 + 4\pi r^2 = 6\pi r^2$. $\frac{dS}{dr} = 12\pi r$. Volume $V = \pi r^2 h = 2\pi r^3$. $\frac{dV}{dr} = 6\pi r^2$, so $\frac{dr}{dV} = \frac{1}{6\pi r^2}$. $\frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dV} \cdot \frac{dV}{dt}$, given that $\frac{dV}{dt} = 12\pi$, $\frac{dS}{dt} = 12\pi r \times \frac{1}{6\pi r^2} \times 12\pi$, $= \frac{24\pi}{r}$. When $r = 4$, $\frac{dS}{dt} = \frac{24\pi}{4} = 6\pi$ (m ² /min).	• Correct expression for $\frac{dS}{dr}$ and $\frac{dr}{dV}$.	• Correct rate of change, with evidence of derivative.	

(e)	Let the height of point B be x , By 45° isosceles triangle and Pythagoras theorem, $AB^2 = x^2 + (1 + x)^2$, $AB = \sqrt{2x^2 + 2x + 1}$. $BC = 20 - 2 - 2x = 18 - 2x$. Total flight time $T = \frac{2AB}{280} + \frac{BC}{200}$, $T = \frac{2\sqrt{2x^2+2x+1}}{280} + \frac{18-2x}{200}$. $\frac{dT}{dx} = \frac{2x+1}{140\sqrt{2x^2+2x+1}} - \frac{1}{100}$. To find minimum time, Set $\frac{dT}{dx} = 0$, $\frac{2x+1}{140\sqrt{2x^2+2x+1}} = \frac{1}{100}$, $7\sqrt{2x^2 + 2x + 1} = 10x + 5$, $x^2 + x - 12 = 0$, $(x - 3)(x + 4) = 0$, $x = 3$ or $x = -4$, but $x \neq -4$ as time cannot be negative. When $x = 3$, $T = \frac{67}{700}$ hours, ≈ 0.0957 hours ≈ 5.74 minutes.	• Finds $\frac{dT}{dx}$, but with one minor error.	• Correct $\frac{dT}{dx}$ AND Sets $\frac{dT}{dx} = 0$.	E7 (T1) Correct x value that minimises time obtained, with evidence of a calculus method OR Correct solution, but with one minor error. E8 (T2) Correct minimum flight time found, with evidence of a calculus method.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Cut Scores (2025)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 13	14 – 19	20 – 24

Cut Scores (2024)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 12	13 – 18	19 – 24